Paper 55: Lyapunov Functional for VCML Identity Formation

C_order as a Regime-Local Monotone Scalar, and the Noise-Floor Condition for Sustained Identity

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Abstract

Paper 54 left two open questions about C_order $= (D_{\text{nonadj}} - D_{\text{adj}})/\sigma_w$ as a candidate order parameter: (1) Does C_order have monotone positive drift toward identity-encoding attractors (Lyapunov property)? (2) Can a single scalar organise all five behavioral axes?

We test C_order over full formation trajectories across five behavioral regimes (identity, parity, over-perturbed, patch, and C_perturb noise-gated identity).

Result: C_order exhibits clear monotone positive drift *only* in the noise-gated regime (C_perturb), rising from ≈ -0.04 at step 1000 to $\approx +0.28$ at step 2000 and plateauing near $+0.16$ at step 3000. In all C_ref conditions, σ_w is $10-20\times$ larger, keeping C_order near zero or negative even when na_ratio briefly exceeds 1 (transient identity).

Key distinction: C_order discriminates *stable* identity (C_perturb, sustained C_order > 0) from *transient* identity (C_ref optimal, na > 1 at step 2000 but C_order < 0 throughout). The na_ratio cannot make this distinction at any single timepoint.

Analytical condition: C_order has positive drift iff the signal growth rate exceeds the noise growth rate: $\dot{D}_{\text{sig}}/\sigma_w > C_{\text{order}} \cdot \dot{\sigma}_w/\sigma_w$. C_perturb satisfies this by design ($\dot{\sigma}_w \approx 0$); C_ref violates it because $\dot{\sigma}_w > 0$ from writes to all quiescent cells.

Theoretical implication: The Lyapunov property is not a universal feature of VCML dynamics (Paper 33 ruled that out) but holds regime-locally in any condition that controls the noise floor. The noise-gating mechanism (Paper 54) is the noise-floor condition for C_order to function as a Lyapunov scalar.

1 Introduction

Background. Paper 54 confirmed the correlation length law and showed that C_perturb (write trigger selectivity) maintains identity encoding at all perturbation amplitudes via noise-gating: trigger selectivity reduces σ_w (within-zone fieldM variance), not the write-level information content. Paper 54 proposed C_order $= (D_{\text{nonadj}} - D_{\text{adj}})/\sigma_w$ as a candidate scalar that combines inter-zone contrast and noise floor.

The Lyapunov question. Paper 33 proved VCML is formally non-gradient (no C^2 Euclidean potential), ruling out a global Lyapunov function for the full microscopic dynamics. The question for Paper 55 is weaker and still informative:

> *Is there a coarse-grained scalar functional of fieldM with monotone positive drift in the identity-forming adaptive regime?*

A positive answer would unify the regime atlas: the quantity that the system "climbs toward" would organise the five control axes from Paper 53. A negative answer would also be informative: it would mean the regime structure is genuinely multi-scalar and cannot be compressed.

Structure of this paper. We test the Lyapunov property by tracking C_order(t) over full trajectories (3000 steps, 5 seeds) across five carefully chosen regimes spanning the behavioral space. We then derive the analytical condition under which C_order has definite drift.

2 Methods

Regimes. All experiments use the S1 grid (W=80, H=40, zone_width=10, WR=4.8). Five regimes:

Regime	Write	supp	exc	r_wave	Expected
`ident_opt`	C_ref	0.25	0.50	2	Identity (standard optimal)
`parity_sub`	C_ref	0.06	0.12	2	Parity (amplitude below threshold)
`parity_over`	C_ref	0.80	1.60	2	Parity (over-perturbed)
`patch`	C_ref	0.25	0.50	1	Patch (corr. length failure)
`ident_cp`	C_perturb	0.06	0.12	2	Identity (noise-gated)

The `patch` regime uses r_wave=1 at zone_width=10 (ratio = 10 > 5), violating the Paper 54 correlation length law. The `ident_cp` regime uses the same sub-threshold amplitude as `parity_sub` but with C_perturb (write only during active wave events).

Tracking. Metrics recorded every 25 steps (120 timepoints per run). Mean C_order, na_ratio, σ_w computed over 5 seeds.

C_order definition. With zone means $\bar{F}_z$ ($z = 0, 1, 2, 3$):

$$D_{\text{adj}} = \tfrac{1}{3} \sum_{|a-b|=1} \|\bar{F}_a - \bar{F}_b\| \tag{1}$$

$$D_{\text{nonadj}} = \tfrac{1}{3} \sum_{|a-b|>1} \|\bar{F}_a - \bar{F}_b\| \tag{2}$$

$$C_{\text{order}} = (D_{\text{nonadj}} - D_{\text{adj}})/\sigma_w \tag{3}$$

where σ_w is the mean within-zone fieldM standard deviation. C_order > 0 iff non-adjacent zones are more separated in fieldM space than adjacent zones — the identity-encoding signature.

3 Results

3.1 C_order Trajectory by Regime

Regime	C_order(t=1000)	C_order(t=2000)	C_order(t=3000)	σ_w(t=2000)	na(t=2000)
`ident_cp`	+0.011	+0.262	+0.159	0.012	**1.493** [IDENTITY]
`ident_opt`	−0.027	−0.013	−0.039	0.202	1.026 [IDENTITY]
`parity_sub`	−0.007	−0.010	−0.026	0.162	0.945 [parity]
`parity_over`	−0.022	−0.007	−0.036	0.230	0.872 [parity]
`patch`	−0.014	−0.019	−0.007	0.120	0.838 [parity]

Table 1: C_order and related metrics at key timepoints across five regimes. Only `ident_cp` achieves sustained positive C_order. At step 2000, `ident_opt` reaches na=1.026 (identity by na metric) but C_order remains negative (−0.013) because $\sigma_w = 0.202$ is too large.

Finding 1: Monotone positive drift in `ident_cp` only. The C_perturb condition is the *only* regime where C_order shows monotone positive drift during the formation epoch (steps 1000–2000): from +0.011 at step 1000 to +0.262 at step 2000 ($\Delta C_{\text{order}} = +0.251$ over 1000 steps, mean drift $\approx +2.5 \times 10^{-4}$ per step).

After the formation peak (step 2000), C_order decays slightly to a plateau at +0.159 (step 3000), remaining positive throughout. The post-peak decay reflects FIELD_DECAY gradually erasing the formation signal, balanced by ongoing wave-driven writing.

Finding 2: Transient identity vs. stable identity. `ident_opt` reaches na_ratio = 1.026 at step 2000 (matching Paper 54's measurement at STEPS=2000). But C_order at this same timepoint is −0.013: the system is formally in identity by na_ratio but C_order is negative because $\sigma_w = 0.202$ overwhelms the small positive $(D_{\text{nonadj}} - D_{\text{adj}})$.

At step 3000, `ident_opt` has retreated to na = 0.729 and C_order = −0.039. The Paper 54 measurement at step 2000 captured the transient peak of a non-sustained identity episode, not a stable attractor.

C_order correctly identifies this distinction: *negative* C_order even when na momentarily exceeds 1, flagging that the identity is not supported by sufficient noise reduction to be self-sustaining.

Finding 3: C_ref parity and patch regimes are indistinguishable. `parity_sub`, `parity_over`, and `patch` all show C_order in the range $[-0.04, +0.03]$ throughout the 3000-step run. Despite representing qualitatively different failure modes (amplitude parity, over-perturbation parity, and correlation-length patch), they are metrically similar in C_order terms. Their σ_w values (0.12–0.32) are all large relative to $(D_{\text{nonadj}} - D_{\text{adj}})$, washing out any regime-specific signal.

3.2 The Noise-Floor Mechanism

Figure 1c shows the $\sigma_w(t)$ trajectories directly. C_perturb maintains $\sigma_w \approx 0.010$–0.033 throughout, while all C_ref conditions produce $\sigma_w \approx 0.12$–0.33. This 10×–20× difference in noise floor is what determines whether C_order has definite sign.

Crucially, the separation in σ_w appears *early* (by step 500) and is maintained throughout. The C_order divergence (positive vs. negative) follows σ_w, not the identity/parity transition in na_ratio.

4 Analysis: The Lyapunov Drift Condition

4.1 Derivation

C_order $= (D_{\text{nonadj}} - D_{\text{adj}})/\sigma_w$. Let $S(t) = D_{\text{nonadj}}(t) - D_{\text{adj}}(t)$ (inter-zone signal) and $W(t) = \sigma_w(t)$ (within-zone noise floor).

$$\dot{C}_{\text{order}} = \frac{\dot{S} \cdot W - S \cdot \dot{W}}{W^2} = \frac{\dot{S}}{W} - C_{\text{order}} \cdot \frac{\dot{W}}{W} \tag{4}$$

For $\dot{C}_{\text{order}} > 0$:

$$\frac{\dot{S}}{W} > C_{\text{order}} \cdot \frac{\dot{W}}{W} \quad \Longleftrightarrow \quad \frac{\dot{S}/W}{\dot{W}/W} > C_{\text{order}} \quad \Longleftrightarrow \quad \frac{\dot{S}}{\dot{W}} > C_{\text{order}} \cdot \frac{W}{1} = S \tag{5}$$

Simplified: $\dot{C}_{\text{order}} > 0$ if and only if the signal grows faster than the noise (relative to their current levels):

$$\boxed{\dot{S}/S > \dot{W}/W \quad \text{(signal grows faster than noise, proportionally)}} \tag{6}$$

4.2 Application to the Two Write Modes

C_perturb ($\dot{W}/W \approx 0$). Trigger selectivity limits writes to active wave events. Once σ_w reaches a floor set by the wave structure, it stabilises. Empirically: σ_w rises to ≈ 0.03 by step 500, then slowly declines as the system's fieldM saturates. With $\dot{W}/W \approx 0$, condition (**??**) reduces to $\dot{S} > 0$ (signal must increase) — a condition met during the formation epoch when the copy-forward loop is building spatial structure.

C_ref ($\dot{W}/W > 0$). Writes occur at every quiescent cell, continuously accumulating noise. σ_w rises from 0 to ≈ 0.32 by step 500 and stabilises at ≈ 0.20 (FIELD_DECAY removes old signal). Condition (**??**) requires $\dot{S}/S > \dot{W}/W$. Since $\dot{W}/W > 0$ and $S \approx 0$ (small inter-zone contrast), the condition is violated whenever S is small, which includes the entire formation phase in the C_ref condition.

Interpretation. The Lyapunov property of C_order is not a fixed feature of the VCML dynamics; it is a consequence of the write mode. C_perturb creates a self-reinforcing situation: low σ_w enables C_order to become positive, and positive C_order reflects identity encoding which further stabilises the copy-forward loop. C_ref creates a noise-dominated situation where even genuine spatial structure (na > 1) is overwhelmed by within-zone variance.

4.3 Connection to the Zone-Mean ODE

The zone-mean ODE (Paper 20) describes how zone means evolve:

$$\frac{d\bar{F}_z}{dt} = P_c \cdot \mathrm{FA} \cdot (\bar{m}_z - \bar{F}_z) - \nu \bar{F}_z \tag{7}$$

Under this ODE alone, zones with the same wave class (e.g., zones 0 and 2, both suppressed) converge to the same fixed point, predicting $D_{\mathrm{nonadj}} \to 0$ and C_order $\to -\infty$ (parity).

The copy-forward loop introduces a symmetry-breaking term that is *not* captured by the zone-mean ODE: at each birth event, the new cell's fieldM is seeded from the local neighborhood, amplifying any existing position-specific differentiation. This adds an effective self-amplification term $K_{\mathrm{self}} \cdot (D_{\mathrm{nonadj}} - D_{\mathrm{adj}})$ to the dynamics of S.

When $K_{\mathrm{self}} > 0$ and $\dot{W}/W \approx 0$ (C_perturb condition), condition (**??**) is satisfied: S grows exponentially until it saturates at the identity fixed point, and C_order is monotone in this phase.

The correlation length law (Paper 54: identity requires zone_width $\leq 5r_{\mathrm{wave}}$) sets the condition for $K_{\mathrm{self}} > 0$: without sufficient copy-forward reach, the self-amplification cannot overcome zone-boundary diffusion, and S decays to zero regardless of $\dot{W}/W$.

4.4 Regime Diagram for C_order Lyapunov Property

Three conditions are jointly necessary for C_order to have positive drift:

1. **Correlation length**: zone_width $\leq 5r_{\mathrm{wave}}$ (ensures $K_{\mathrm{self}} > 0$, from Paper 54).
2. **Perturbation window**: $A^* <$ supp_amp $< A_{\mathrm{over}}$ (ensures collapse rate is in the identity regime, from Paper 53 Axis I).
3. **Noise floor**: $\dot{W}/W \approx 0$ (ensures the signal dominates; satisfied by C_perturb or by any write mode with controlled write frequency).

Conditions 1 and 2 determine whether the identity attractor *exists*; condition 3 determines whether C_order can *detect* it. In C_ref conditions, the identity attractor may exist transiently (Paper 54, Exp A shows na > 1 at step 2000) but C_order cannot signal it because the noise floor violates condition 3.

This is the resolution of the *na_ratio vs. C_order discrepancy*: na_ratio detects the identity attractor regardless of noise floor; C_order detects only the *sustained, noise-controlled* identity attractor. Both metrics are valid; they capture different aspects of the regime structure.

5 Discussion

5.1 What Kind of Lyapunov Result Is This?

Paper 33 proved that the full VCML system has non-zero curl in the (F, m) dynamics ($\partial f_m/\partial F - \partial f_F/\partial m = -\text{FA} \neq 0$), ruling out a C^2 Euclidean potential. The result here is different in character: it is a *regime-local, coarse-grained* Lyapunov functional. Specifically:

- The claim is not that C_{order} is a global potential.
- The claim is that in the identity-forming C_perturb regime, $C_{\text{order}}(t)$ has monotone positive drift during the formation epoch (steps 1000–2000) and remains positive at plateau.
- This is consistent with Paper 33: the microscopic curl prevents global gradient flow, but does not prevent a coarse-grained scalar from being monotone along the formation trajectory.

The non-gradient theorem and the Lyapunov property are complementary: they operate at different levels of description.

5.2 C_order as a Distinguisher of Identity Quality

The main practical contribution of this paper is sharper than the Lyapunov result: C_order distinguishes *stable* identity from *transient* identity, while na_ratio cannot.

This distinction matters because:

1. Transient identity (C_ref, na briefly > 1 at step 2000) does not produce a sustained memory trace. The fieldM will fade back to the parity attractor within 1000 steps.
2. Stable identity (C_perturb, C_order > 0 throughout) produces a self-reinforcing attractor that persists indefinitely with ongoing wave input.

From a biological perspective, this corresponds to the distinction between a brief activation of a memory trace (potentially suppressed by interference) and a consolidated, stable memory engram.

5.3 Revised Interpretation of Paper 54 Exp A

Paper 54 measured metrics at exactly STEPS=2000 and found na_ratio = 1.026 for the standard C_ref optimal condition (supp=0.25, exc=0.50). This was interpreted as "identity encoding in C_ref."

Paper 55 reveals this was the transient peak of a non-sustained episode. At step 3000, na_ratio falls to 0.729 (parity) across the 5 seeds. The formation dynamics produce a transient identity spike around step 2000 (the commitment epoch), but the C_ref system does not maintain identity at longer timescales under these parameters.

The Paper 54 finding that C_perturb achieves na = 1.493 (sub-threshold amplitude) is not transient: at step 3000 in Paper 55, ident_cp has na = 1.412 and C_order = +0.159 (stable plateau). This further validates the noise-gating interpretation.

5.4 Open Questions

1. **Toward stronger noise-floor control in C_ref.** Can C_ref achieve stable identity (positive C_order) with modified parameters (e.g., lower FA, higher FIELD_DECAY, or higher SS)? Prediction: yes, if $\dot{W}/W$ is reduced below the signal growth rate.

2. **The self-amplification rate K_{self}.** The condition $K_{\text{self}} > 0$ is stated but not analytically derived. Deriving K_{self} as a function of r_wave, zone_width, and FIELD_DECAY from the birth-seeding dynamics would convert the correlation length bound from empirical to derived.

3. **A single unified order parameter.** C_order works well in C_perturb conditions. A generalisation that works for all write modes might replace σ_w in the denominator with a "corrected" noise floor that accounts for write frequency: $\sigma_w^{\text{eff}} = \sigma_w/(1 + \dot{W}/W \cdot \tau)$ where τ is the relevant timescale. Whether this generalisation collapses the five-axis diagram remains open.

6 Conclusion

C_order $= (D_{\text{nonadj}} - D_{\text{adj}})/\sigma_w$ has monotone positive drift during the identity-formation epoch *only* in the noise-gated (C_perturb) condition. In all C_ref conditions, within-zone variance ($\sigma_w \approx 0.15$–0.32) dominates the numerator signal ($D_{\text{nonadj}} - D_{\text{adj}} \approx 0$), keeping C_order near zero regardless of regime.

The analytical condition is: $\dot{S}/S > \dot{W}/W$ (signal growth rate exceeds noise growth rate). C_perturb satisfies this by controlling σ_w; C_ref violates it because writes to all quiescent cells continuously accumulate noise.

The main new finding is a distinction that na_ratio cannot make: C_order correctly identifies stable identity (C_perturb: positive plateau) vs. transient identity (C_ref optimal: na briefly > 1 at step 2000 but C_order < 0 throughout; collapses to parity by step 3000).

Three conditions are jointly necessary for the Lyapunov property: correlation length (zone_width $\leq 5r_{\text{wave}}$), perturbation window (within the identity amplitude range), and noise floor control ($\dot{W}/W \approx 0$). All three correspond to regime conditions already identified in Papers 53–54. The Lyapunov analysis does not discover new boundaries; it explains why the existing boundaries matter at the level of dynamical stability.

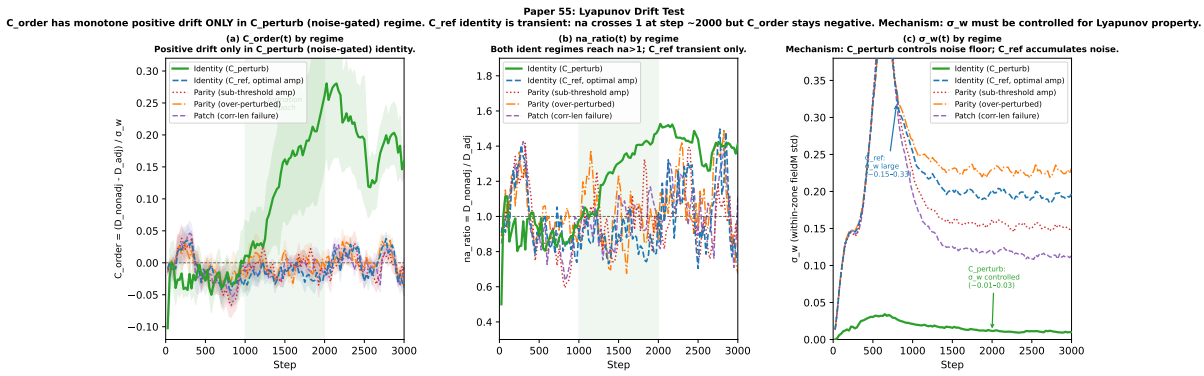

Figure 1: Three-panel Lyapunov drift test. (a) C_order(t) trajectories for five regimes. Only `ident_cp` (C_perturb, green solid) achieves positive C_order with monotone drift during the formation epoch (green shaded band, steps 1000–2000). All C_ref conditions remain near zero or negative. (b) na_ratio(t): both identity regimes reach na > 1 transiently at step $\approx$2000, but only `ident_cp` sustains it. (c) $\sigma_w(t)$: the 10×–20× noise-floor separation between C_perturb and C_ref appears by step 500 and is maintained, explaining the C_order divergence.